

PFMEA vs DFMEA - Key Differences

AIAG-VDA 2019 | Quality Head Agents

Purpose

DFMEA (Design FMEA) analyses potential failure modes in the product design, while PFMEA (Process FMEA) analyses potential failure modes in the manufacturing process. DFMEA is owned by Engineering; PFMEA is owned by Manufacturing Quality.

When to Use

DFMEA: During product design (APQP Phase 2). PFMEA: During process design (APQP Phase 3). Both must be completed before the production trial run and PPAP submission.

Rating Differences

DFMEA uses RPN (Risk Priority Number = $S \times O \times D$). PFMEA now uses Action Priority (AP = H/M/L) per AIAG-VDA 2019. The shift from RPN to AP eliminates the problem of different combinations giving the same RPN with different risk levels.

Key Inputs

DFMEA: Customer requirements, engineering specifications, DVP&R. PFMEA: Process Flow Diagram (PFD), engineering drawings, previous PFMEA, customer complaints, lessons learned.

Linkage to Control Plan

PFMEA must be aligned with the Process Flow Diagram (PFD) - every process step in the PFD must appear in the PFMEA. The PFMEA then drives the Control Plan. This three-way linkage is a critical IATF audit requirement.

IATF Clauses

DFMEA: Clause 8.3.3.3 | PFMEA: Clause 8.5.6.1 | Both: Clause 8.5.1.1 | Control Plan linkage: Clause 8.5.1